

Faculty of Computing

IT1160 – Discrete Mathematics

Year 1 Semester 2 (2026)

Tutorial 03

1. What is the term a_7 of the sequence $\{a_n\}$ if a_n equals:
 - a) $n^2 + 3n$
 - b) 6
 - c) $1 - (2n)$
 - d) $(-3)^n$
2. For each of the following integer sequences, identify a rule or pattern that generates the terms. Based on the pattern, determine the next three terms of each sequence.
 - a) 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, ...
 - b) 1, 3, 9, 27, 81, 243, 729, 2187, 6561, 19683, ...
3.
 - a) Write the general formula for the n th term of an arithmetic sequence where the first term is 5 and the common difference is 3.
 - b) Find the 10th term of the sequence: 3, 7, 11, 15, ... And find the summation of the first 10 terms of the sequence.
 - c) If the 5th term of an arithmetic sequence is 20 and the 10th term is 40, find the first term and common differences.
 - d) How many terms are there in the arithmetic sequence 2, 5, 8, ..., 50?
4.
 - a) If the first term of a geometric sequence is 5 and the common ratio is 2, write the formula for the n th term.
 - b) Find the 6th term of the sequence: 2, 6, 18, 54, And find summation of first 10 terms of the sequence.
 - c) How many terms are there in the geometric sequence: 2, 6, 18, ..., 1458?
 - d) If the 3rd term of a geometric sequence is 24 and the 6th term is 192, find the first term and the common ratio.

5. A cloud storage provider tracks how its storage usage changes over time. At the start of monitoring, the storage usage is **5 GB**. Each month, the storage usage increases by **4 GB** compared to the previous month.

The storage usage after each month follows a regular pattern based on the information above.



- Find storage usage for the **first 15 months**, starting from month 1.
- Calculate the **total storage usage** over these 15 months.
- Use the following formula to independently calculate the total storage usage and **verify** the result obtained in part (b):

$$\text{Sum formula } S_n = \frac{n}{2}(2a + (n - 1)d)$$

where:

- a is the initial storage usage.
- d is the monthly increase.
- n is the number of months.

6. A server processes data packets in **discrete and equally spaced time intervals**. During each time interval, the number of packets processed by the server depends on the interval number. The total number of packets processed over the **first 5-time intervals** is represented by the following expression: $\sum_{i=1}^5 (3i + 2)$ (where i represents the time interval number).
- Write out the expression by **expanding the summation**, showing the number of packets processed in each of the first 5-time intervals.
 - Using the expanded expression, **calculate the total number of packets processed** over these 5-time intervals.
 - Explain, in words, what the **index i** and the **upper limit** represent in the context of this server system.
7. A secure authentication system generates **one-time passwords (OTPs)** using a sequence of independent security checks. At stage i , the system allows **$(i+1)$** possible valid choices for the OTP character. If the OTP generation process consists of **5 stages**, the total number of possible OTPs can be modeled using a product.
- Write the total number of possible OTPs using **product notation**.
 - Expand the product and compute the total number of possible OTPs.
 - Briefly explain why a product (instead of a sum) is used in this situation.

8. A cloud service provider is deploying an application across multiple servers. For each deployment stage i , there are i different configuration options available. The deployment process consists of **6 stages**, and each stage must choose exactly one configuration option.
- Model the total number of distinct deployment configurations using **product notation**.
 - Evaluate the product to find the total number of configurations.
 - State how this result relates to the concept of **factorials**.
9. A computer system generates unique access codes by arranging a set of distinct characters. Each access code is formed by using all the available characters exactly once, and the order of the characters matters. For example, changing the order of the characters results in a different access code.
- If the system uses 5 distinct characters, write the mathematical expression that represents the total number of different access codes that can be generated.
 - Evaluate this expression to determine how many unique access codes can be produced using the 5 characters.
 - In the context of counting and computing, explain why the value of $0! = 1$ is logically defined in mathematics and computer science.
10. A cybersecurity analyst is testing an **intrusion detection system (IDS)** by simulating a set of **distinct attack patterns**. Each attack pattern must be launched **exactly once**, and the **order in which the attacks are executed matters**, since different sequences may trigger different system responses. If there are **6 distinct attack patterns**,
- Write a mathematical expression using **factorial notation** to represent the total number of possible attack sequences.
 - Evaluate the expression to find the total number of different attack sequences.
 - State why factorial is an appropriate mathematical model for this situation.